EXPLICIT COMPUTATIONS OF THE ADDITIVE CONSTANTS

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Supplementary material to *Operator Fejér–Riesz Theorems with Gaps and Degree Bounds* [JZ26]. This document provides the detailed computations referred to in the remark at the end of the subsection "Computation of the additive constant" in the manuscript.

We use the notation of [JZ26]: $\mathcal{B}(\mathcal{H})$ denotes the bounded operators on a Hilbert space $\mathcal{H}$, and $\mathrm{SOS}_{\mathcal{H}}$ denotes the finite sums of hermitian squares of polynomials in $\mathcal{B}(\mathcal{H})[x]$. For a finite set $S \subset \mathbb{R}[x]$, a certificate in the operator quadratic module $\mathcal{M}_{\mathcal{H}}(S)$ has the form

$$F = \sum_{g \in S \cup \{1\}} g\,\sigma_g, \qquad \sigma_g \in \mathrm{SOS}_{\mathcal{H}}\,.$$

Its weighted degree is the maximum of $\deg(g\sigma_g)$ over the nonzero summands; the subscript $\leq n$ means that such a certificate exists with weighted degree at most n. We write $M(S)$ for the scalar quadratic module and $\mathrm{SOS}_{\mathbb{C}}$ for the scalar sums of hermitian squares. References of the form [JZ26, Theorem 3.4] refer to the manuscript; the claims and equations below are numbered within this document.

1. Explicit examples

The following example illustrates the computation of the additive constants obtained in the proof of [JZ26, Theorem A]. We first consider a union of two disjoint compact intervals and then examine how adjoining an isolated point affects the resulting degree bound. The accompanying Mathematica notebook is available in the repository [JZ26-code]. The constants computed below are admissible bounds; we do not assert their optimality. The minimality statements concern the specific intermediate certificates in Claims 2, 3, and 5.

Example 1.1. Let

$$K_0 = [0,1] \cup [2,3], \qquad K_1 = K_0 \cup \{4\},$$

with natural descriptions

$$S_0 = \{x,\ (x-1)(x-2),\ 3-x\},$$
$$S_1 = \{x,\ (x-1)(x-2),\ (x-3)(x-4),\ 4-x\}.$$

By [JZ26, Theorem 2.2], both $M(S_0)$ and $M(S_1)$ are saturated. The union of the nondegenerate interval components is K_0 in both cases. Thus $\ell = 2$, and the associated endpoint polynomials are

$$P = x(x-2), \qquad Q = (x-1)(x-3), \qquad R = 3-2x, \qquad \Delta = -x(x-1)(x-2)(x-3).$$

The associated rank-one pencil is $T(y) = D - yvv^{\mathsf{T}}$, where

$$D = \mathrm{diag}(b_1, b_2) = \mathrm{diag}(1, 3),$$

and the entries of v may be chosen positive, with $v_j^2 = \frac{P(b_j)}{Q'(b_j)}$, $j = 1, 2$. In the present example, $v_1^2 = \frac{1}{2}$, $v_2^2 = \frac{3}{2}$, so we can choose

$$v = \begin{pmatrix} 1/\sqrt{2} \\ \sqrt{3/2} \end{pmatrix}.$$

Claim 1. The recovery multiplier $w(x) = v_1^2(x-b_2)^2 + v_2^2(x-b_1)^2$ satisfies

$$w(x) = 2\left((x-\tfrac{3}{2})^2 + \tfrac{3}{4}\right).$$

In particular, $w > 0$ on $\mathbb{R}$.

Proof of Claim 1. Since $v_1^2 = \frac{1}{2}$ and $v_2^2 = \frac{3}{2}$, we have

$$w(x) = \frac{1}{2}(x-3)^2 + \frac{3}{2}(x-1)^2 = 2\left((x-\tfrac{3}{2})^2 + \tfrac{3}{4}\right),$$

which proves the claim. □

Claim 2. There exist $\sigma_0, \sigma_1 \in \mathrm{SOS}_{\mathbb{C}}$ satisfying

$$\Delta = \sigma_0 + w\Delta\sigma_1,$$

with $\deg \sigma_1 = 4$. Moreover, no such identity is possible with $\deg \sigma_1 < 4$.

Proof of Claim 2. Set

$$\sigma_1(x) = \frac{1}{4}\left((x-\tfrac{3}{2})^2 - \frac{19}{12}\right)^2 + \frac{1}{18},$$
$$\tau(x) = \frac{1}{2}(x-\tfrac{3}{2})^2 + \frac{1}{24},$$

and $\sigma_0 = \Delta^2\tau$. Direct expansion gives

$$1 = w\sigma_1 + \Delta\tau,$$

and hence

$$\Delta = \sigma_0 + w\Delta\sigma_1.$$

Clearly, $\sigma_0, \sigma_1 \in \mathrm{SOS}_{\mathbb{C}}$ and $\deg \sigma_1 = 4$.

We now show that the degree 4 of σ_1 is the smallest possible among all SOS identities of the form $\Delta = \sigma_0 + w\Delta\sigma_1$. Suppose that

$$\Delta = \widetilde{\sigma}_0 + w\Delta\widetilde{\sigma}_1, \qquad \widetilde{\sigma}_0, \widetilde{\sigma}_1 \in \mathrm{SOS}_{\mathbb{C}}.$$

Since Δ changes sign at $0, 1, 2, 3$, we must have

$$w(x)\widetilde{\sigma}_1(x) = 1 \qquad (x = 0, 1, 2, 3).$$

The unique polynomial of degree at most three satisfying these conditions is

$$p(x) = \frac{1 + 3x - x^2}{6}.$$

However, $p(x) < 0$ for sufficiently large $|x|$, so $p \notin \mathrm{SOS}_{\mathbb{C}}$. Consequently, no such identity is possible with $\deg \widetilde{\sigma}_1 < 4$. □

By Claim 2 and [JZ26, Theorem 4.8], the quadratic certificate bound for $\mathcal{M}_{\mathcal{H}}(\{\Delta\})$ is

$$4\ell - 2 + \deg \sigma_1 = 6 + 4 = 10.$$

We next transfer this bound to the natural description S_0.

Claim 3. The polynomial Δ admits a certificate in $M(S_0)$ of weighted degree 7, and no certificate of smaller weighted degree exists.

Proof of Claim 3. A computation verifies that

$$\begin{aligned}\Delta = \frac{1}{6}x\big[(x-1)(x-2)(x-3)\big]^2 \\ + \frac{1}{2}(x-1)(x-2)\big[x(x-3)\big]^2 + \frac{1}{6}(3-x)\big[x(x-1)(x-2)\big]^2.\end{aligned} \tag{1.1}$$

Each multiplier is a scalar sum of squares, and the largest weighted degree is 7. Thus $\Delta \in M(S_0)_{\leq 7}$.

To prove minimality, suppose that Δ admits a certificate of weighted degree at most 6. Write

$$\Delta = \tau_0 + x\tau_1 + (x-1)(x-2)\tau_2 + (3-x)\tau_3,$$

where $\tau_j \in \mathrm{SOS}_{\mathbb{C}}$.

At each zero of Δ, every summand is nonnegative. Consequently, whenever a generator is strictly positive at such a zero, its SOS multiplier must vanish there. Since a nonnegative real polynomial has only zeros of even multiplicity, every nonzero multiplier must satisfy $\deg \tau_0 \geq 8$, $\deg \tau_1 \geq 6$, $\deg \tau_2 \geq 4$, and $\deg \tau_3 \geq 6$. Indeed, these bounds correspond to the required double zeros at the endpoints where the respective generators are strictly positive.

A certificate of weighted degree at most 6 therefore forces $\tau_0 = \tau_1 = \tau_3 = 0$. The remaining multiplier must have the form $\tau_2 = c\,x^2(x-3)^2$, $c \geq 0$. Hence

$$\Delta = c(x-1)(x-2)x^2(x-3)^2.$$

If $c > 0$, the right-hand side has degree 6, whereas $\deg \Delta = 4$. If $c = 0$, the identity is also impossible. Thus no certificate of weighted degree at most 6 exists. Together with (1.1), this proves that the minimum weighted degree is 7. □

By Claim 3, the minimum weighted degree of a certificate for Δ in $M(S_0)$ is 7. Hence, in [JZ26, Corollary 4.9], we may take $\eta_{S_0} = 7 - 2\ell = 3$. Combining this with Claim 2 gives the quadratic bound $N_{S_0} = 10 + 3 = 13$. Therefore, by [JZ26, Theorem 3.4], every hermitian $F \in \mathcal{B}(\mathcal{H})[x]$ with $\deg F \leq \mathtt{d}$, $\mathtt{d} \geq 1$, and $F \succeq 0$ on K_0 satisfies

$$F \in \mathcal{M}_{\mathcal{H}}(S_0)_{\leq 11+2\lceil \mathtt{d}/2 \rceil} \subseteq \mathcal{M}_{\mathcal{H}}(S_0)_{\leq \mathtt{d}+12}.$$

Thus, we obtain the admissible additive constant $c_{S_0} = 12$.

Adjoining the isolated point. We now pass from K_0 to $K_1 = K_0 \cup \{4\}$. In the notation of [JZ26, Section 5], the isolated point is $\xi = 4$, and

$$\delta = -\Delta(4) = 24.$$

The matrix polynomial used to adjoin this point is

$$G_4(x) = \begin{pmatrix} 24 + \Delta(x) & (x-4)\Delta(x) \\ (x-4)\Delta(x) & (x-4)^2\Delta(x) \end{pmatrix}.$$

For this single isolated point, we may choose the interpolation polynomial $\psi_1 = 1$ in [JZ26, Proposition 5.4]. Then $a = 1 - \psi_1^2 = 0$, so it suffices to find a matrix certificate for G_4 in $\mathcal{M}_{\mathbb{C}^2}(S_1)$.

Claim 4. The matrix polynomial G_4 admits a certificate in $\mathcal{M}_{\mathbb{C}^2}(S_1)$ of weighted degree 9.

Proof of Claim 4. Define polynomials

$$P_1 = (x-1)(x-2)(x-3)(x-4), \qquad P_2 = x(x-3)(x-4),$$
$$P_3 = x(x-1)(x-2), \qquad P_4 = x(x-1)(x-2)(x-3),$$

and set

$$\tau = x^2 - 3x + \frac{27}{8}, \qquad \rho_1 = \frac{1}{4}(4-x)^2\tau, \qquad \rho_4 = x^2 + \frac{3}{2} + \frac{1}{4}x^2\tau.$$

Define the polynomial columns

$$u_3 = \begin{pmatrix} -\frac{1}{4} \\ \frac{1}{6}P_3 \end{pmatrix}, \qquad u_4 = \begin{pmatrix} \frac{3}{4} \\ \frac{1}{6}P_4 \end{pmatrix}.$$

With these choices, direct expansion gives

$$\begin{aligned} G_4 = x\begin{pmatrix}\rho_1 & 0\\ 0 & \frac{1}{6}P_1^2\end{pmatrix} + (x-1)(x-2)\begin{pmatrix}0 & 0\\ 0 & \frac{1}{2}P_2^2\end{pmatrix} \\ + 6(x-3)(x-4)u_3u_3^{\mathsf{T}} + (4-x)\left(6u_4u_4^{\mathsf{T}} + \begin{pmatrix}\rho_4 & 0\\ 0 & 0\end{pmatrix}\right). \end{aligned} \tag{1.2}$$

To verify that this is an SOS certificate, observe that $\tau = (x-\frac{3}{2})^2 + \frac{9}{8} > 0$ on $\mathbb{R}$. Therefore, $\rho_1, \rho_4 \in \mathrm{SOS}_{\mathbb{C}}$. The two rank-one matrices $u_3u_3^{\mathsf{T}}$ and $u_4u_4^{\mathsf{T}}$ are also hermitian squares. Thus every matrix multiplier in (1.2) belongs to $\mathrm{SOS}_{\mathbb{C}^2}$. Finally, inspection of the degrees in (1.2) shows that the largest weighted degree is 9. Hence $G_4 \in \mathcal{M}_{\mathbb{C}^2}(S_1)_{\leq 9}$, as claimed. □

The choice $\psi_1 = 1$ therefore gives an additive increase of at most 9 in [JZ26, Proposition 5.4]. We now show that allowing a different interpolation polynomial cannot reduce this degree within the construction.

Claim 5. For every real polynomial ψ satisfying $\psi(4) = 1$, the matrix polynomial $\psi^2 G_4$ admits no certificate in $\mathcal{M}_{\mathbb{C}^2}(S_1)$ of weighted degree at most 8. Consequently, the minimum weighted degree achievable for this fixed matrix construction is 9.

Proof of Claim 5. Suppose, for a contradiction, that $\psi(4) = 1$ and $\psi^2 G_4$ has a certificate of weighted degree at most 8. Taking the $(2,2)$ entry gives a scalar certificate in $M(S_1)$ for

$$f(x) = \psi(x)^2(x-4)^2\Delta(x).$$

Since $\deg f = 6 + 2\deg\psi$, the assumed weighted degree bound forces $\deg\psi \leq 1$. Write the resulting scalar certificate as

$$f = \tau_0 + x\tau_1 + (x-1)(x-2)\tau_2 + (x-3)(x-4)\tau_3 + (4-x)\tau_4,$$

where every $\tau_j \in \mathrm{SOS}_{\mathbb{C}}$. The polynomial f vanishes at $0, 1, 2, 3, 4$. As in Claim 3, at any such point where a generator is strictly positive, the corresponding SOS multiplier must have a double zero. The resulting minimum degrees for nonzero multipliers are $\deg\tau_0 \geq 10$, $\deg\tau_1 \geq 8$, $\deg\tau_2 \geq 6$, $\deg\tau_3 \geq 6$, and $\deg\tau_4 \geq 8$. Since the certificate has weighted degree at most 8, we must have $\tau_0 = \tau_1 = \tau_4 = 0$. Moreover, the remaining two multipliers must be of the form

$$\begin{aligned} \tau_2 &= a\,x^2(x-3)^2(x-4)^2, \\ \tau_3 &= b\,x^2(x-1)^2(x-2)^2, \end{aligned} \qquad a, b \geq 0.$$

Substituting these expressions into the scalar certificate for f, and using $\Delta = -x(x-1)(x-2)(x-3)$, we obtain

$$\begin{aligned} -\psi^2 x(x-1)(x-2)(x-3)(x-4)^2 = a\,x^2(x-1)(x-2)(x-3)^2(x-4)^2 \\ + b\,x^2(x-1)^2(x-2)^2(x-3)(x-4). \end{aligned}$$

Cancelling the common polynomial factor gives

$$-\psi^2(x-4) = a\,x(x-3)(x-4) + b\,x(x-1)(x-2). \tag{1.3}$$

We distinguish the two possible degrees of ψ.

If ψ is constant, then $\psi(4) = 1$ implies $\psi = 1$. Evaluating (1.3) at $x = 0$ gives $4 = 0$, a contradiction.

If $\deg\psi = 1$, the leading coefficient of the left-hand side of (1.3) is strictly negative, whereas the leading coefficient of the right-hand side is $a + b \geq 0$. This is again impossible.

Thus no interpolation polynomial ψ with $\psi(4) = 1$ can give a certificate of weighted degree at most 8. Together with Claim 4, this establishes the asserted minimum degree 9. □

We can therefore apply [JZ26, Proposition 5.4] with $\eta_{S_1} = 9$. Combining this with the quadratic bound 10 from Claim 2 gives $N_{S_1} = 10 + 9 = 19$. Therefore, for every hermitian $F \in \mathcal{B}(\mathcal{H})[x]$ with $\deg F \le \mathtt{d}$, $\mathtt{d} \ge 1$, and $F \succeq 0$ on K_1, [JZ26, Theorem 3.4] gives

$$F \in \mathcal{M}_{\mathcal{H}}(S_1)_{\le 17+2\lceil \mathtt{d}/2 \rceil} \subseteq \mathcal{M}_{\mathcal{H}}(S_1)_{\le \mathtt{d}+18}.$$

Thus, we obtain the admissible additive constant $c_{S_1} = 18$.

Declaration of AI use

Large language models assisted in working out these examples by applying methods developed in [JZ26]. The authors have independently verified all mathematical arguments and take full responsibility for the content and conclusions of this supplementary material.

References

[JZ26] A. Jindal and A. Zalar, *Operator Fejér–Riesz Theorems with Gaps and Degree Bounds*, manuscript, 2026.

[JZ26-code] A. Jindal and A. Zalar, *Operator Fejér–Riesz Theorems with Gaps and Degree Bounds: supplementary examples and verification code*, https://github.com/ZalarA/Operator-Fejer-Riesz-theorems.